

Difference-in-Differences

SW Chapter 10 (Part 2)

ECON3500: Econometrics and Applications

Spring 2026

Quick Review: What We Covered Tuesday

Last class (with Rae), we covered:

Types of data: cross-sectional, time-series, panel, repeated cross-sections

First differencing: $\Delta x_t = x_t - x_{t-1}$ — subtracting periods eliminates time-invariant α_i

Fixed effects: give each entity its own intercept (α_i)

- Equivalent to demeaning: estimating deviations from each unit's mean
- **Within** estimator (FE) vs. **between** estimator (pooled OLS)

Today: a specific research design built on these tools — **difference-in-differences**.

The Problem with Simple Comparisons

Motivating Example: Garbage Incinerator

Question: What is the effect of a garbage incinerator on nearby housing prices?

After the incinerator was built:



$$\widehat{rprice} = 101,308 - 30,688 \cdot nearinc$$

Houses near the incinerator sell for ~\$30k less. Did the incinerator cause this?

Not So Fast...

Look at the relationship **before** the incinerator was built:

$$\widehat{rprice} = 82,517 - 18,824 \cdot nearinc$$

The incinerator was built in a place where housing prices were **already depressed!**

The \$30k gap reflects both the incinerator effect **and** pre-existing differences.

Two Flawed Comparisons

⚠️ Cross-Sectional Comparison (After Only)

Compare near vs. far houses **after** the incinerator: $-\$30,688$

Problem: Near-incinerator houses were **already cheaper**. Location characteristics are **confounders** — they affect both proximity to the incinerator *and* housing prices.

⚠️ Before/After Comparison (Treatment Group Only)

Compare near-incinerator houses before vs. after: $+\$18,790$

Problem: Housing prices were **rising everywhere**. We're mixing the treatment effect with a common time trend.

We need a method that handles **both** problems simultaneously.

Difference-in-Differences

The Core Logic



Subtract out the pre-existing difference:

$$\hat{\delta}_1 = \underbrace{(-30,688)}_{\text{after gap}} - \underbrace{(-18,824)}_{\text{before gap}} = -11,864$$

The incinerator reduced nearby prices by ~\$12k — not \$30k.

The 2x2 Table



The **difference of differences**:

	Before	After	
Control (far)	\$82,517	\$101,308	+\$18,790
Treatment (near)	\$63,693	\$70,619	+\$6,927
Treat – Control	–\$18,824	–\$30,688	–\$11,864

$$\underbrace{(70,619 - 63,693)}_{\Delta \text{ treatment}} - \underbrace{(101,308 - 82,517)}_{\Delta \text{ control}} = [-11,864]. \textit{alert}$$

DiD Graphically

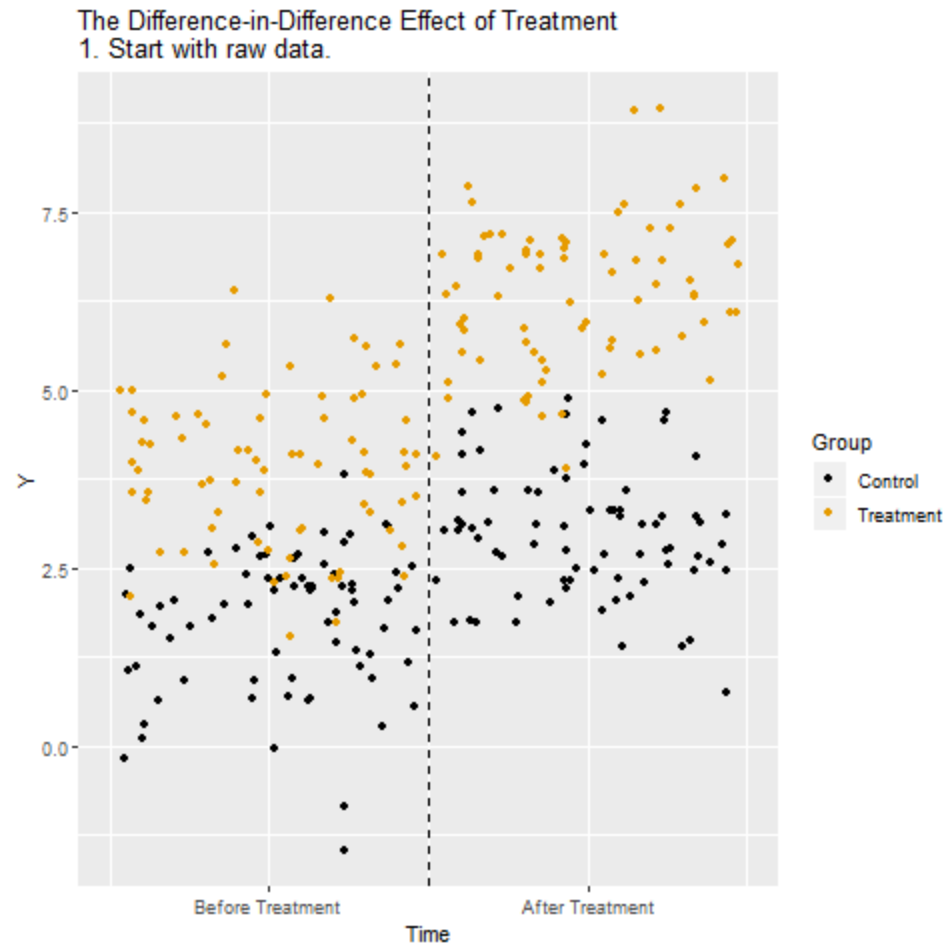
The dashed line shows where near-incinerator prices *would have been* if they followed the same trend as far-away prices.

The **DiD estimate** is the vertical gap between:

What **actually happened** to the treatment group

The **counterfactual**: what would have happened without treatment

DiD Animated



Animation by Nick Huntington-Klein ([The Effect](#))

Watch the steps:

1. Raw data: two groups over time
2. Collapse to group means
3. Measure the control group's time trend
4. **Subtract** that trend from the treatment group
5. What remains = the **DiD estimate**

The key: use the control group's trajectory to build the **counterfactual** for the treatment group.

The Counterfactual

! What DiD Really Estimates

DiD doesn't compare treatment and control groups directly. It asks:

*How much did the treatment group change **relative to what would have happened without treatment?***

The control group's trajectory provides the counterfactual — but only if parallel trends holds.

This is why DiD is so useful for **policy evaluation**:

Treatment and control groups can have **very different levels**

They just need to be on **similar trajectories** before the policy hits

The identifying assumption is about **trends**, not **levels**

The Regression Framework

DiD as a Regression

We can capture the entire DiD logic in one regression:

$$rprice_i = \beta_0 + \delta_0 \cdot after_i + \beta_1 \cdot nearinc_i + \delta_1 \cdot (after_i \times nearinc_i) + u_i$$

	Before (α)	After (α)	Change
Control (far)	β_0	$\beta_0 + \delta_0$	δ_0
Treatment (near)	$\beta_0 + \beta_1$	$\beta_0 + \delta_0 + \beta_1 + \delta_1$	$\delta_0 + \delta_1$
Treat – Control	β_1	$\beta_1 + \delta_1$	δ_1

δ_1 — the coefficient on the **interaction term** — is the DiD estimate.

Interpreting Each Piece

$$y_i = \underbrace{\beta_0}_{\text{baseline}} + \underbrace{\delta_0 \cdot after_i}_{\text{time trend}} + \underbrace{\beta_1 \cdot treated_i}_{\text{group difference}} + \underbrace{\delta_1 \cdot (after_i \times treated_i)}_{\text{treatment effect}} + u_i$$

Coefficient	Meaning
β_0	Average outcome for control group, before
δ_0	How the control group changed over time (common time trend)
β_1	Pre-existing difference between groups
δ_1	The DiD estimate — causal <i>if</i> parallel trends holds

Adding Control Variables

Why add controls to a DiD regression?

1. **Precision:** Controls **reduce residual variance** → tighter standard errors, even if parallel trends holds unconditionally.
2. **Credibility:** Parallel trends may only hold **conditional on covariates**. If treatment and control groups differ on observables that predict trends, controlling for those variables makes parallel trends more plausible.

Works fine — just add $\mathbf{X}_{it}$ to the regression:

$$y_{it} = \beta_0 + \delta_0 \cdot after_t + \beta_1 \cdot treated_i + \delta_1 \cdot (after_t \times treated_i) + \gamma' \mathbf{X}_{it} + u_{it}$$

Controls Can Matter for Identification

Unlike in an RCT, parallel trends may only hold *after conditioning* on observables. Omitting those variables is **OVB**.

Example: If newer homes were built farther from the incinerator site during this period, controlling for house characteristics isn't just about precision — it's about identification.

The Parallel Trends Assumption

The Key Assumption

! Parallel Trends Assumption (*common trends assumption in SW*)

In the **absence of treatment**, the difference between treatment and control groups would have remained **constant over time**.

Equivalently: both groups would have followed **parallel trajectories**.

Key point: Parallel trends is a statement about what **would have happened without treatment** — it is about the untreated counterfactual, not about what we observe after treatment.

What this **allows**:

Treatment and control groups can have **different levels** of the outcome

There can be **time-invariant** unobserved confounders — DiD differences them out

What this **requires**:

No **time-varying confounders** that differentially affect the two groups

Parallel Trends Holds

Groups follow the same trajectory before treatment. The treatment effect is the gap between the actual outcome and the counterfactual (dashed line).

The dashed line = the **counterfactual**: where the treatment group would have been if it followed the control group's trend.

Parallel Trends Violated

Groups were converging before treatment. DiD attributes the continued convergence to the treatment — overstating the true effect.

The gold dotted line = the **true counterfactual** (continuing convergence). The dashed line = what DiD **incorrectly assumes**. The difference between the two is the bias.

What Would Violate Parallel Trends?

In the incinerator example:

A new highway built near the incinerator site at the same time

A neighborhood revitalization program targeting the area

Differential migration (people leaving *because* the incinerator was announced)

More generally, any factor that:

Changes over time (not absorbed by entity or time FE)

Affects treatment and control groups **differently**

Coincides with the timing of treatment

Assessing Parallel Trends

We can never **prove** parallel trends — it's about what **would have happened** without treatment.

But we can look for **supporting evidence**:

1. Pre-treatment trend test: Plot the outcome for both groups over time. Do they move in parallel *before* treatment?

2. Placebo/falsification tests:

Run DiD using a **fake treatment date** (before the real one). A significant “effect” suggests something else is driving the result.

Run DiD on an outcome that **should not be affected** by treatment.

3. Event study plot: Estimate treatment effects for each period relative to treatment. Pre-treatment coefficients should be **near zero and flat**.

The Event Study Plot

An **event study** estimates separate effects for each period relative to treatment:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \cdot D_{it}^k + u_{it}$$

where $D_{it}^k = 1$ if unit i is k periods from treatment at time t .

Think of it as DiD estimated separately for each time period — revealing *when* effects emerge and whether they grow, fade, or were already present before treatment.

What to look for:

Pre-treatment coefficients ($k < 0$): Should be close to zero → supports parallel trends

Post-treatment coefficients ($k \geq 0$): Show the dynamic treatment effect over time

The reference period ($k = -1$) is normalized to zero

Why Event Studies Matter

Event study plots have become nearly **mandatory** in applied DiD papers. They are the most credible way to support the parallel trends assumption.

Real-World Example

Card & Krueger (1994): Does the Minimum Wage Kill Jobs?

Question: Does raising the minimum wage reduce employment?

Setting:

New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992

Pennsylvania did not change its minimum wage

Surveyed fast-food restaurants in both states, before and after

	Before	After	Δ
NJ (treatment)	20.44 FTE	21.03 FTE	+0.59
PA (control)	23.33 FTE	21.17 FTE	-2.16
NJ – PA			+2.76

Result: Employment **increased** in NJ relative to PA — the opposite of the standard prediction. One of the most influential papers in labor economics.

Why Is Card & Krueger Compelling?

What makes this a good DiD design?

Sharp treatment: NJ raised the minimum wage on a specific date; PA didn't

Geographic neighbors: NJ and PA share economic conditions → plausible parallel trends

Same industry: Fast food in both states faces similar demand shocks

No obvious differential shocks: No major PA-specific event in 1992 that would have changed fast-food employment

What might you be concerned about?

Were NJ restaurants **anticipating** the wage increase? (announcement effects violate sharp timing)

Did PA have its own employment shocks unrelated to the minimum wage?

Are the two states similar *enough* for parallel trends?

Measurement error: Employment data came from phone surveys of managers

DiD in Stata

Estimating DiD

Option 1: Interaction regression

```
* Generate interaction term
gen after_treat = after * treated

* Estimate DiD
reg y after treated after_treat, robust
```

Option 2: Factor variable notation (preferred)

```
* Stata creates the interaction automatically
reg y i.after##i.treated, robust
```

The coefficient on `1.after#1.treated` is the DiD estimate $\hat{\delta}_1$.

DiD with Panel Data in Stata

With true panel data (same entities over time), combine DiD with fixed effects:

```
* Set panel structure
xtset entity_id year

* Entity + time FE with DiD
xtreg y treated_post i.year, fe vce(cluster entity_id)
```

Or equivalently, using `reghdfe` :

```
reghdfe y treated_post, absorb(entity_id year) vce(cluster entity_id)
```

Don't Forget to Cluster!

With panel data, **always** cluster standard errors at the entity level — we'll cover exactly why on Tuesday.

Wrapping Up

DiD and Internal Validity

Ch 9 Threat	How DiD Helps	What DiD Cannot Fix
OVB	Eliminates time-invariant confounders by differencing	Time-varying confounders that differentially affect groups
Wrong functional form	Flexible — can add covariates, nonlinearities	Misspecified treatment timing or group definitions
Measurement error	Not addressed	Attenuation bias still applies
Sample selection	Not addressed	Differential attrition (people move <i>because</i> of treatment)
Simultaneous causality	Treatment timing helps	If treatment is <i>responsive</i> to anticipated outcomes

DiD Checklist

Before you trust a DiD estimate, ask:

- ☐ Is there a clear **treatment group** and **control group**?
- ☐ Is there a clear **before** and **after** period?
- ☐ Is **parallel trends** plausible? What evidence supports it?
- ☐ Are there **pre-trend tests** or an **event study plot**?
- ☐ Could any **time-varying confounder** have changed differentially at the same time?
- ☐ Are standard errors **clustered** appropriately? *(we'll discuss why on Tuesday)*

Key Takeaways

DiD combines a **cross-sectional** comparison with a **time** comparison — the causal interpretation requires **parallel trends**

The **interaction term** captures the DiD estimate ($\hat{\delta}_1$)

Parallel trends is **untestable** — it's about a counterfactual — but pre-trends and event studies provide supporting evidence

DiD addresses **time-invariant OVB** by differencing it out — but **time-varying confounders** that differentially affect groups still threaten validity

Always **cluster standard errors** at the entity level with panel data — more on this Tuesday